

This is a classic CI/CD workflow setup

Step 1: Created a Simple Bash Script File that prints system info:

```
touch script.sh; vi script.sh
```

```
# !/bin/bash  
echo "Build started..."  
echo "System Info:"
```

```
uname -a
```

```
echo "Build completed successfully!"
```

Make the script executable:

```
chmod +x script.sh
```

```
md_rustam@DESKTOP-CPK0PUB:~/Project$ cd 16_Jenkins_Task_2/  
md_rustam@DESKTOP-CPK0PUB:~/Project/16_Jenkins_Task_2$ vi script.sh  
md_rustam@DESKTOP-CPK0PUB:~/Project/16_Jenkins_Task_2$ chmod +x script.sh
```

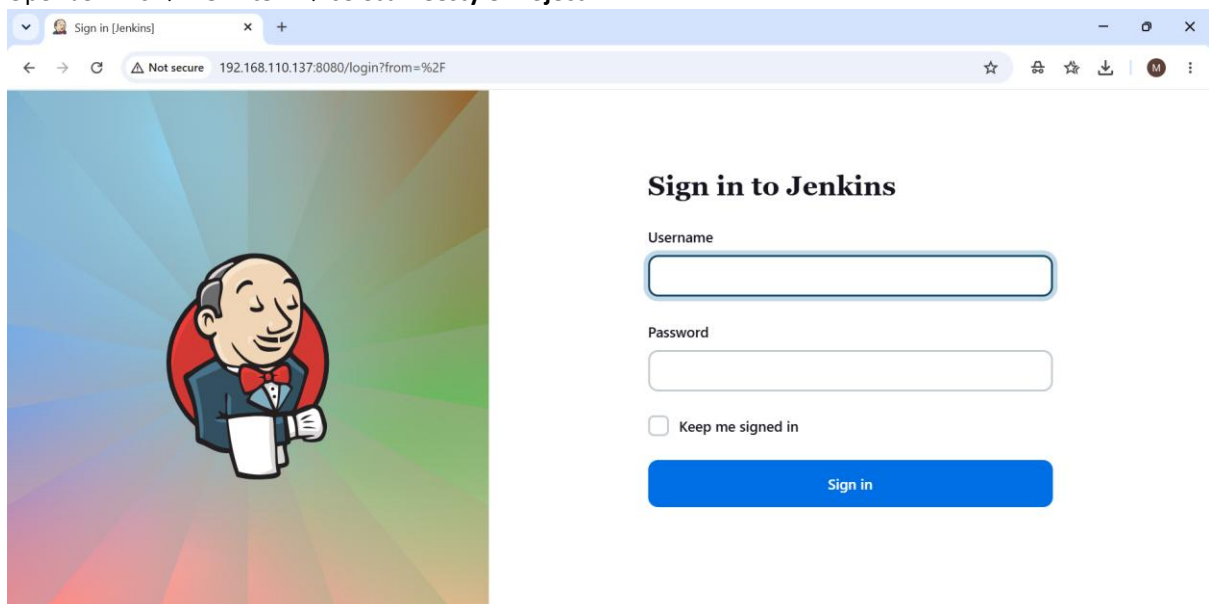
Commit and push to the GitHub repo.

```
git add script.sh; git commit -m "Add build script"; git push origin main
```

Step 2: Configure Jenkins Project (Open Jenkins → **New Item** → select **Freestyle Project**):

```
$ systemctl status jenkins  
● jenkins.service - Jenkins Continuous Integration Server  
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset:  
enabled)  
   Active: active (running) since Sun 2026-08-02 08:44:51 UTC; 4min  
10s ago  
   Main PID: 737 (java)  
     Tasks: 45 (limit: 4544)  
    Memory: 1.1G (peak: 1.1G)  
       CPU: 1min 31.309s  
    CGroup: /system.slice/jenkins.service  
            └─737 /usr/bin/java -Djava.awt.headless=true -jar /usr/  
share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080
```

Open Jenkins → **New Item** → select **Freestyle Project**



Sign in to Jenkins

Username

Password

☐ Keep me signed in

Sign in

← → ↻ ⚠ Not secure 192.168.110.137:8080/view/all/newJob ☆ ⚙ ⭐ ⬇

Dashboard > All >

Enter an item name

GitHub-Auto-Build

» Required field

**Freestyle project**

Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

If you want to create a new item from other existing, you can use this option:

**Copy from**

Type to autocomplete

OK

Under **Source Code Management**, choose **Git** and paste your repo URL. Under **Build Triggers**, *GitHub hook trigger for GITScm polling*

▼ GitHub-Auto-Build #4 Console × +

← → ↻ ⚠ Not secure 192.168.110.137:8080/job/GitHub-Auto-Build/4/console ☆ ⚙ ⭐ ⬇

**Jenkins**

Q Search (CTRL+K) ? 🔔

Dashboard > GitHub-Auto-Build > #4 > Console Output

 Status

 Changes

 Console Output

 View as plain text

 Edit Build Information

 Delete build '#4'

 Previous Build

 **Console Output** 

Started by user unknown or anonymous
Running as SYSTEM
Building in workspace /var/lib/jenkins/workspace/GitHub-Auto-Build
Finished: SUCCESS

<http://192.168.110.137:8080/github-webhook/>

**Jenkins**

Q Search (CTRL+K) ? 🔔 3 🛡 3

Dashboard > Nodes >

 Nodes

 Clouds

Build Queue ^

Build Executor Status v

1 Idle

2 Idle

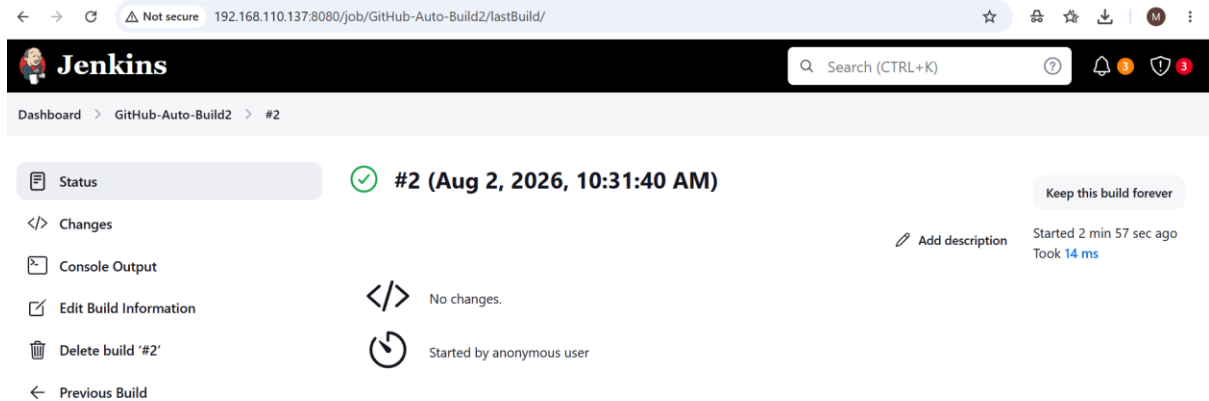
Nodes

+ New Node Configure Monitors ↻

S	Name	Architecture	Clock Difference	Free Disk Space	Free Swap Space	Free Temp Space	Response Time
	Built-In Node	Linux (amd64)	In sync	3.60 GiB	1.95 GiB	3.60 GiB	0ms
Data obtained		7 min 57 sec	7 min 57 sec	7 min 57 sec	7 min 57 sec	7 min 57 sec	7 min 57 sec

Icon: S M L Legend

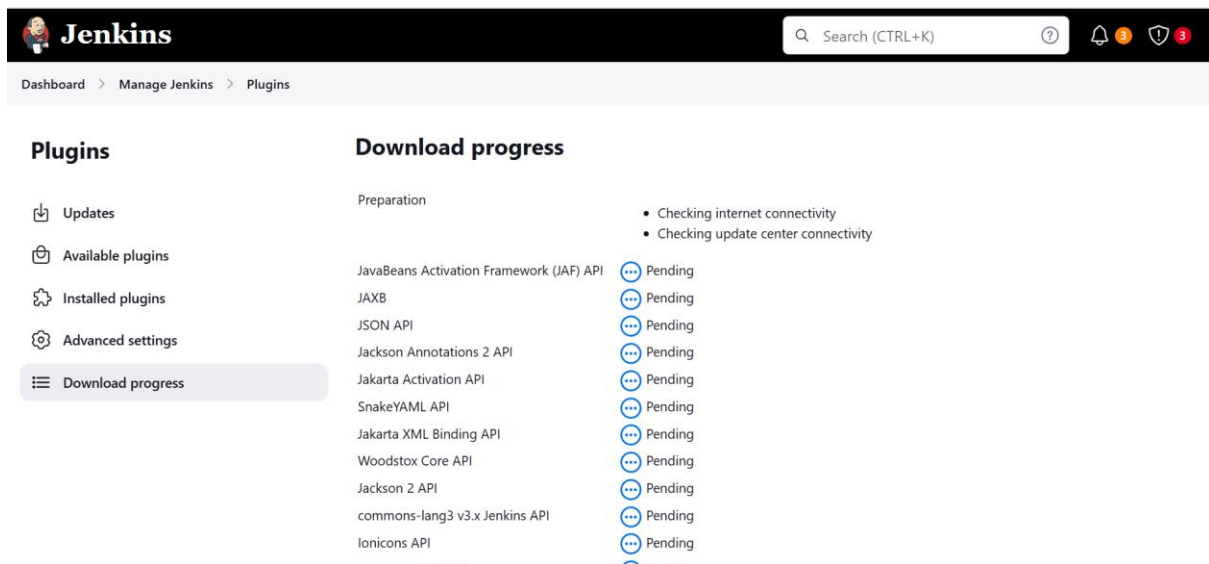
Step 3: Configure GitHub Webhook, Add Build Step, In Jenkins job → Build Steps → Execute shell:



The screenshot shows the Jenkins job status page for job #2. The top navigation bar includes the Jenkins logo, a search bar, and notification icons. The breadcrumb trail is 'Dashboard > GitHub-Auto-Build2 > #2'. The main content area shows the job status as 'Success' with a green checkmark and the build number '#2 (Aug 2, 2026, 10:31:40 AM)'. A 'Keep this build forever' button is visible. On the left, there are links for 'Changes', 'Console Output', 'Edit Build Information', 'Delete build #2', and 'Previous Build'. The 'Changes' section shows 'No changes.' and 'Started by anonymous user'.

Email Notification

1. Install Email Extension Plugin in Jenkins.



The screenshot shows the Jenkins 'Plugins' page. The left sidebar contains links for 'Updates', 'Available plugins', 'Installed plugins', 'Advanced settings', and 'Download progress'. The 'Download progress' section is active, showing a list of plugins being downloaded. The 'Download progress' table lists the following plugins and their status:

Plugin Name	Status
Preparation	Checking internet connectivity, Checking update center connectivity
JavaBeans Activation Framework (JAF) API	Pending
JAXB	Pending
JSON API	Pending
Jackson Annotations 2 API	Pending
Jakarta Activation API	Pending
SnakeYAML API	Pending
Jakarta XML Binding API	Pending
Woodstox Core API	Pending
Jackson 2 API	Pending
commons-lang3 v3.x Jenkins API	Pending
Ionicons API	Pending

2. Configure SMTP in Manage Jenkins → Configure System:

- Go to **Manage Jenkins** → **Configure System**.
- Scroll down to **Extended E-mail Notification**.
- Fill in the SMTP details:

3. smtp.gmail.com

4. Port: 587

5. Use TLS